

ATLAS OFFENSIVE · VERSION 1.0 · RETURN FIRST

MASTER RUN BOOK

The complete, self-contained program of research runs for building a high-return systematic investment machine — with the backstory any LLM needs to implement each run correctly.

Prepared 21 July 2026 · 66 run cards across 9 stages (R00–R63 + R2B/R2C) · every run marked
ADD / UPGRADE / REPLACE-IF-BETTER

Prime directive: highest defensible net compound return. Risk metrics recorded, not gated, until Stage 7.

One evolving portfolio — the Offensive Book — connects every run. Failure is output; the bench remembers everything.

Production pin untouched: **improved_frontier_phase5_fragility_guard**

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PART A — BACKSTORY: READ THIS BEFORE ANY RUN

A.1 What this project is and how we got here

Atlas Allocation is a weekly ETF tactical allocation system built over ~60 research phases. Architecture: Layer 1 alpha signals (momentum, trend, reversal, breadth, dollar-strength, macro/VIX features, all lagged 1 week) → Layer 2A sleeves (8 sleeve strategies) → Layer 2B causal 5-state regime engine (`stressed_panic`, `recovery_fragile`, `recovery_confirmed`, `neutral_mixed`, `calm_trend`) → Layer 3 HRP portfolio construction with regime multipliers, target-vol scaling, a BIL (T-bill) defensive overlay, and a crowding guardrail.

Production result: **7.13% CAGR, 0.948 Sharpe, -11.6% max drawdown, SPY beta 0.24, average 27.6% cash**. Over the same period SPY compounded ~10.5%. The system is an excellent *defensive* product — and that is exactly the problem this program exists to solve.

A.2 The five lessons that shape every run below

Lesson 1 — The universe was the ceiling, not the ideas. 35 collinear ETFs behave like ~5 independent bets. Every cross-sectional idea (residual momentum, reversal, carry/value, ML rankers, latent factor discovery) failed or plateaued *on this universe* while the same ideas have strong published evidence *on wide stock universes*. Runs that move to single stocks are not retrying failures; they are running the experiment for the first time.

Lesson 2 — Long-only strangled the relative signals. Track C proved reversal/residual-momentum signals have real information (holdout rank IC up to 0.407 — the best signal evidence in the repo) yet lose money expressed as long-only ETF tilts. Long/short expression is not optional for these families.

Lesson 3 — The defense mandate suppressed measured, validated return. The Moonshot episode map: in 76 early-recovery weeks SPY compounded at +73% annualized while the system held 53% cash and earned 12%. The four biggest opportunity gaps in 21 years are all early recoveries (-23pp, -15pp, -12pp, -10pp per episode vs the achievable bound). 67% of those weeks are labeled `stressed_panic`, where every overlay multiplier is frozen at 1.0 *by convention*. A validated fix (PBI, below) is capped at +0.003 Sharpe purely by that convention. Separately, the production signal amplitude ($\alpha=0.08$) is provably 3–6× under-sized — walk-forward selection picks 0.24–0.40 at essentially every checkpoint since 2009, and what stops it is a turnover-cost gate, not information.

Lesson 4 — Models never beat features here. A leakage-controlled kNN analog engine found real signal and still only *matched* the hand-built composite (IC 0.090 vs 0.092). GBM, trust models, meta-allocators, learned states — all failed on ~1,000 weekly samples. ML earns its place only after breadth exists (millions of panel observations). Never run heavy ML before the data layer that feeds it.

Lesson 5 — The options verdicts are void. All four internal options experiments (v1 REJECT, v2 REJECT, recovery RESEARCH-ONLY, v3 REJECT) priced options with a Black-Scholes proxy on realized vol — a fiction every report itself flags. Options are an *untested* domain, not a rejected one.

A.3 The return-first doctrine (what “we deal with other metrics later” means operationally)

1. **Primary metrics in every run:** net CAGR, expected log growth, per-trade expectancy, profit factor, and residual alpha after regressing on SPY + standard factors. Rank candidates by these.
2. **Recorded but non-gating:** volatility, max drawdown, CVaR, Sharpe, turnover, concentration. Report them honestly; never reject a candidate for them during discovery.

3. **Integrity gates (always binding):** point-in-time data, survivorship-free universes, 1-period signal lags, time-ordered splits, per-instrument realistic costs (never the old 10bps flat), every tested variant logged in the trial registry, null controls (shuffled/inverted/random-placement) for any headline claim, and the sealed 2026+ holdout untouched.
4. **A “beta is not alpha” rule with a twist:** Track B taught that more return via more beta is trivially available. We still *report* the decomposition (beta vs alpha) in every run — but under the return-first doctrine, *fuller intelligent beta is an acceptable win* (e.g., cutting the 27.6% cash drag) as long as it is labeled as what it is. We just refuse to *call* it alpha.
5. **Risk engineering is Stage 7, not a reflex.** Stops, vol caps, tail hedges, and drawdown budgets are added at the end, priced explicitly (each protection must state how much CAGR it costs). Never re-impose the old 10% vol target, BIL floors, or Sharpe-first gates inside a discovery run.

A.4 Standing implementation rules for any LLM executing a run

- Read `CLAUDE.md` first. Never modify production code, pins, weights, or `public/dashboard-data.json`. Never `git add -A`; stage files by name.
- Each run works in its own namespace: `scripts/atlas_offensive_runNN/` with outputs in `data/research/atlas_offensive_runNN/`, and ends with a report in `docs/research/atlas_offensive/runNN_report.md`.
- Pre-register before running: hypothesis, candidates, parameter grids, primary metric, success/failure thresholds — written to a file *before* the first backtest. Post-hoc discoveries are allowed but must be labeled post-hoc and re-confirmed in a locked follow-up before being believed (this is how the α /PBI findings were handled — copy that pattern).
- Every run report answers: What was run? What improved *return*? Standalone and in combination with the current best book? Verdict: **Adopt into book** / **Confirm in follow-up** / **Research-only** / **Drop permanently**. Plus: trial count, warnings, git status.
- The stack of record for combination tests is the **Offensive Book** (defined in A.5), not the defensive production pin. The pin is a reporting benchmark only.
- If a prerequisite (data subscription, capital decision, prior run’s artifact) is missing: **stop and report**, don’t improvise substitutes.

A.5 The Offensive Book — the single thread connecting all runs

Every run either **adds an engine**, **upgrades a component**, or **replaces a component** of one evolving portfolio called the **Offensive Book**. This is how runs “correlate and keep improving”:

- **Book v0** = 100% of offensive capital in the best simple benchmark (SPY buy-and-hold). Any adopted run must beat or diversify this, net of costs.
- Each run card below states **Builds on** (which prior runs/artifacts it consumes) and **Improves or Replaces** (which Book component it upgrades or swaps out).
- After every adopted run, re-run the standing **Book integration test**: does the Book’s net CAGR / log growth improve when the new component is added at its proposed weight? A component that wins standalone but not in combination is held in the bench registry, not deleted.
- The Book’s component slots: `EQUITY_CORE` (long stock exposure engine), `REGIME_BRAIN` (state machine deciding aggression), `LS_ALPHA` (long/short sleeve), `TREND_MACRO` (futures sleeve), `OPTIONS` (convexity/premium sleeve), `SHORT_HORIZON` (overnight/daily sleeve), `EVENT` (event-driven sleeve), `SIZING` (position sizing/leverage logic), `ALLOCATOR` (capital allocation across sleeves), `DEFENSE` (the old Atlas system, kept as an optional capital-preservation slot — deliberately last).

A.6 Data purchases the program assumes (buy when the run needing it starts)

Purchase	~Cost	First needed by
Norgate US stocks (PIT constituents, survivorship-free, delistings)	~\$110/mo	Run 1
Sharadar SF1 fundamentals (25y PIT as-reported)	~\$50/mo	Run 6
ORATS or ThetaData options chains (2007+)	~\$100–160/mo	Run 13
Norgate futures or Databento CME (micro futures)	~\$100–200/mo	Run 8
Polygon or Databento intraday bars	~\$125–199/mo	Run 17

Free and already sufficient: FRED/ALFRED vintages, SEC EDGAR (Form 4/13F/filings), FINRA short interest, CBOE indices, crypto exchange APIs.

PART B — PROGRAM MAP

B.1 Stages

Stage	Runs	Theme	Book effect
0 Foundation	R00–R01	Governance + first data unlock	Creates the rules and the breadth signal
1 Internal unlocks	R02, R2B, R2C	Un-freeze the regime engine	Builds REGIME_BRAIN v1
2 Breadth platform	R03–R07	Single-stock universe, momentum→factors→long/short	Builds EQUITY_CORE + LS_ALPHA
3 New markets	R08–R12, R17–R21	Futures, macro, carry, events, earnings, overnight, alt-data, crypto	Builds TREND_MACRO, EVENT, SHORT_HORIZON
4 Options	R13–R16	The untested domain, on real data	Builds OPTIONS
5 ML ladder	R22–R34	GBM workhorse + challengers	Upgrades EQUITY_CORE/LS_ALPHA signal quality
6 Math transfers	R35–R41	Bounded cross-field experiments	Feature/component upgrades anywhere
7 Integration & risk	R42–R47	Sizing, leverage, allocation, then (and only then) drawdown engineering	Builds SIZING + ALLOCATOR, prices DEFENSE
8 Evergreen	R48–R63	Refresh, expand, regenerate — the program that never runs out	Continuous Book improvement

B.2 The improve-vs-replace convention

Every card carries one of: - **[ADD]** — creates a new Book component (empty slot filled). - **[UPGRADE]** — improves an existing component; old version stays as fallback in the bench registry. - **[REPLACE-IF-BETTER]** — head-to-head against an existing component; winner takes the slot, loser goes to the bench with its results. - **[FEEDER]** — produces signals/data/infrastructure consumed by other runs, no direct Book slot. - **[PROBE]** — cheap feasibility test that decides whether a family of runs opens or closes.

B.3 Critical path and parallel tracks

Critical path: R00 → R01 → R03 → R04 (first deployable Book v1) → R05 → R22 → R42 → R43 → R44 → R45 → R47. **Parallel any time after R00:** futures track (R08→R09/R10/R21), options track (R13→R14/R15/R16), short-horizon track (R17→R18), free alt-data (R19). **Rule:** at most one heavy run in flight per track; cheap probes may interleave.

PART C — THE RUN CARDS

Template: **Backstory** (why this run exists — what we know), **Mission** (return-first question), **Builds on / Improves-Replaces**, **Do** (implementation sketch), **Success / Failure / Next**.

STAGE 0 — FOUNDATION

R00 — Governance Reset [FEEDER] · difficulty 2 · \$0 · days

Backstory. Atlas's greatest asset is its honesty culture: walk-forward labels, pre-registration, null controls. Its old holdout (2024-04-19) is burned from repeated use, and its old gates were Sharpe-first — structurally hostile to return-seeking. Both must be rebuilt before anything else, or every later result is untrustworthy. **Mission.** Stand up the return-first rulebook every subsequent run obeys. **Builds on:** existing Phase D validation stack. **Improves:** replaces Sharpe-first gate set with the A.3 doctrine. **Do.** (1) Seal new holdout: declare a 2026 cutoff date; data after it untouched until final Book promotion. (2) Create trial registry (one table: run_id, variant, params, primary metric, verdict) — every variant logged. (3) Write pre-registration template. (4) Build per-instrument cost library: measured ETF/stock spreads, futures commissions+rolls, options at-spread fills, borrow schedule; retire the 10bps flat model. (5) Codify the A.3 return-first metric set as the standard report table. **Success:** registry + templates exist; first cost library committed. **Failure:** n/a — skipping this is the only failure. **Next:** R01 immediately; R08/R13/R17 may start in parallel.

R01 — PIT Stock Breadth Confirmation [FEEDER→UPGRADE of REGIME_BRAIN] · difficulty 2 · ~\$110/mo · 1–2 weeks

Backstory. The single best-validated internal diagnostic: stock-level breadth predicted +0.517% per 4 weeks in calm markets where ETF-level breadth showed -0.457%. But it was measured on survivorship-biased current constituents. The pipeline (`scripts/build_pit_stock_breadth_panel.py`) is already built and waiting for point-in-time data. Calm markets are 27% of all weeks and the system currently earns ~4.2% annualized in them while SPY runs away — this is the cheapest known attack on the biggest passive gap. **Mission.** Does the breadth lift survive honest data? If yes, how much offense can it justify in calm states? **Builds on:** R00; Phase 5A scaffold. **Improves:** REGIME_BRAIN's calm-state aggression logic. **Do.** Buy Norgate. Export PIT S&P 500 membership + delisting-aware prices to 2005. Build `pct_above_200d`, advance/decline, net new highs (weekly, lagged). Reproduce the 5A diagnostic unbiased. If lift $\geq$ half the biased number, test breadth-scaled offense (pre-registered tiers $\times 0.85/1.00/1.15$, then a bolder $\times 1.0/1.15/1.30$ return-first grid) in calm and neutral states. **Success:** unbiased lift confirmed → adopt breadth as a REGIME_BRAIN input; feeds R02/R03. **Failure:** lift $<$ half → breadth demoted to diagnostic; Norgate still justified by R03. **Next:** R02 and R03 in parallel.

STAGE 1 — UN-FREEZE THE REGIME ENGINE

R02 — Offensive Regime Rebuild: PBI Native [UPGRADE of REGIME_BRAIN] · difficulty 5 · \$0 · 3–6 weeks

Backstory. The most important suppressed finding in the repo. “Panic-but-improving” (PBI): inside deep-drawdown panic weeks, three lagged confirmations (credit improving, 4-week breadth change positive, VIX term structure back in contango) identify improving panic. It fired in 9 distinct episodes over 21 years, beat 91% of random-placement nulls, its inverted version *hurts* (-0.025 — direction is real), it improved stressed-panic performance, and a locked confirmation sprint passed it three ways. Yet its measured effect is $\sim +0.003$ Sharpe — because the production wrapper freezes panic offense at $\sim 15\%$ by convention. Early-recovery weeks are where SPY does $+73\%$ annualized. This run removes the freeze. **Mission.** How much CAGR is recoverable when confirmed-improving panic is allowed real offense? **Builds on:** R01 (breadth confirmation input); Moonshot/Confirm1 artifacts. **Improves:** REGIME_BRAIN — adds a sixth state `stressed_panic_improving`; the frozen-modifier convention is **replaced**. **Do.** (1) Implement PBI as a native Layer-2B sub-state (reuse the locked Confirm1 rule: deep-DD latch $\leq -10\%$ within 13w; 2-of-3 confirmations $\times 1.15$, 3-of-3 $\times 1.30$ — then extend with pre-registered offense-base grids 25/40/55/70% for the sub-state itself). (2) Hard per-episode stop: if drawdown deepens $\geq X\%$ after re-risk, revert to panic defense (grid $X \in \{3, 5, 7\}\%$). (3) Benchmark the whole state engine against a statistical jump model (transition-penalized clustering) and an HMM for label stability. (4) Replay 2008 explicitly — the improving-then-collapsing panic is THE failure mode. **Success:** average early-recovery episode capture $+5\text{pp}$ or better, 2008 replay survivable $\rightarrow$ REGIME_BRAIN v2 adopted; feeds R15 (same trigger buys calls). **Failure:** stop-losses can’t contain 2008-type reversals $\rightarrow$ keep PBI at confirmed modest amplitude (the Confirm1 version) and log the ceiling. **Next:** R2B, R2C.

R2B — Amplitude Liberation (α Re-Derivation) [UPGRADE of REGIME_BRAIN] · difficulty 3 · \$0 · 1–2 weeks

Backstory. The production offense-scaling signal (R2A state quality) runs at $\alpha=0.08$, a value pre-declared conservatively in 2026 and never revisited. Walk-forward objective selection — no lookahead — picks α at the top of the grid at essentially every checkpoint since 2009. Higher α passed all old gates up to 0.40, but under the *old* objective it traded CAGR for Sharpe ($7.13\% \rightarrow 6.97\%$). That trade was an artifact of the defensive base: with cash floors, scaling “offense” mostly de-risked earlier. On an offensive base the same signal should scale exposure up, not down. **Mission.** Re-derive the optimal amplitude of state-quality scaling under a log-growth objective on the offensive base (post-R02 REGIME_BRAIN, reduced cash floor). **Builds on:** R02. **Replaces:** the $\alpha=0.08$ convention. **Do.** Walk-forward α selection with expected-log-growth as the objective; cash floor 0–10% except unconfirmed panic; report the full α curve (return, growth, drawdown) rather than a single point; shuffled-signal null at chosen amplitude (the Moonshot control showed the timing is real — verify it still is on the new base). **Success:** growth-optimal α adopted into REGIME_BRAIN v2. **Failure:** signal value doesn’t survive the base change $\rightarrow$ document and keep state scaling modest. **Next:** R2C.

R2C — Macro Sub-States for the Dead Zone [UPGRADE of REGIME_BRAIN] · difficulty 5 · \$0 · 3–6 weeks

Backstory. `neutral_mixed` is 44% of all weeks and is a no-conviction catch-all. Internal macro-quadrant work (growth $\times$ inflation factors from FRED data) found a real $+1.41\%/4\text{w}$ development-period spread — the `slowdown` sub-regime inside `neutral_mixed` returned $+1.47\%/4\text{w}$ vs $+0.24\%$ for `stress` — but holdout rank order inverted, and weekly-overlay expressions all failed. Diagnosis: vintages weren’t used (revised data leaked), lags were too optimistic, and monthly information was forced into weekly decisions. **Mission.** Split the 44% dead zone into actionable sub-states using honestly-lagged vintage macro data at monthly cadence.

Builds on: R02; V1–V3 classifier code. **Improves:** REGIME_BRAIN's neutral_mixed handling. **Do.** Rebuild the PCA quadrant classifier on ALFRED *vintage* data (free) with a 2-month publication lag; expanding-window, sign-anchored; quadrants condition *monthly offense budgets* within neutral_mixed only; explicitly test whether the 2024–26 “stress-but-rallying” holdout anomaly persists under vintages (it may have been a data artifact). **Success:** holdout rank consistency + neutral_mixed return improvement → adopt as monthly conditioning layer. **Failure:** quadrants unstable under honest lags → close the macro-overlay family permanently (it will have had three honest chances). **Next:** Stage 2 continues regardless.

STAGE 2 — THE BREADTH PLATFORM (EQUITY_CORE + LS_ALPHA)

R03 — Single-Stock Universe + Momentum Family [ADD: EQUITY_CORE foundation] · difficulty 5 · in Norgate · 1–2 months

Backstory. Lesson 1 in action. Cross-sectional momentum is the most replicated return source in finance (93 countries, a century of data; survives the “replication crisis” audit), and Atlas has never actually tested it — dual-momentum over 35 ETFs is ~5 effective bets, not a cross-section. The panel built here (PIT membership, delisting returns, per-name costs) is the platform every later equity run inherits: this is the program’s single most important infrastructure investment. **Mission.** Build the panel; establish which momentum variants deliver net return on 500–2,000 stocks. **Builds on:** R01 (Norgate live). **Feeds:** every equity run (R04–R07, R11–R12, R19–R20, R22+). **Do.** (1) Panel: Russell-1000-like PIT universe, daily prices, delistings, splits/dividends, per-name spread estimates. (2) Variants: 12-1, 6-1, 52-week-high proximity, residual momentum (vs rolling 3-factor fits), vol-scaled, sector-relative. (3) Measure decile spreads (informational) and implementable top-k long portfolios (deployable). (4) Decade-by-decade decomposition; the planning assumption is post-publication decay of $\frac{1}{3}$ – $\frac{1}{2}$ — post-2010 results decide. **Success:** ≥ 1 variant with post-2010 net top-k return > SPY → EQUITY_CORE candidate signals exist. **Failure:** nothing beats SPY net → EQUITY_CORE falls back to intelligent-beta (R04 still runs — concentration + regime gating of plain exposure). **Next:** R04 (deploy), R05 (short leg), R06 (factors).

R04 — Concentrated Compounder: Book v1 [ADD: EQUITY_CORE] · difficulty 4 · in Norgate · 2–4 weeks

Backstory. Atlas’s HRP allocator spreads capital across everything, always — conviction was never allowed to concentrate. Under return-first, concentration is the cheapest known monetizer of signal strength (breadth × IC arithmetic). This run creates the first deployable offensive product: a 5–20 name portfolio from R03’s best ranks, run with the REGIME_BRAIN deciding aggression. **Mission.** Beat SPY’s compound return net of costs with a concentrated, regime-aware, long-only stock portfolio. **Builds on:** R03 ranks + R02 REGIME_BRAIN. **Improves-Replaces:** Book v0 (SPY) → Book v1; also replaces HRP-style dilution for this sleeve. **Do.** Grid $k \in \{5, 10, 20\}$, rebalance $\in \{1w, 2w, 1m\}$; sector cap 40%; per-position trailing stops (recorded, since stops are also a return decision); REGIME_BRAIN scales gross exposure 0.7–1.3. Compare directly against the same signals HRP-diluted (the concentration test old Atlas never ran). **Success:** net CAGR > SPY + 2pp in walk-forward → **Book v1 adopted.** **Failure:** concentration adds turnover and vol but not return → $k=20+$ diversified variant becomes EQUITY_CORE. **Next:** R05, R06 upgrade its signals; R42 later re-sizes it.

R05 — Long/Short Equity v1 [ADD: LS_ALPHA] · difficulty 7 · in Norgate · 2–3 months

Backstory. Lesson 2. Track C’s residual momentum and reversal sleeves had genuine predictive content and still lost money long-only — the short leg is where half (often more) of relative-signal return lives. Nobody has ever shorted a share in this project. The open question isn’t the signals; it’s whether retail shorting economics (borrow fees, locates, margin) leave the spread intact. **Mission.** Does a dollar-neutral book on our best relative signals make money net of real borrow/costs? **Builds on:** R03 panel + signals. **Adds:** LS_ALPHA slot. **Do.** Easy-borrow large-cap universe only; dollar-neutral, $|\text{beta}| < 0.1$ verified, sector-neutralized; signals: residual momentum + short-term reversal + (post-R06) factor composites; collect live IBKR borrow fees from day one and haircut history with them; cost-doubling stress mandatory; squeeze filter via short-interest screens. **Success:** net market-neutral return > 5% annualized (report Sharpe, don’t gate on it) → LS_ALPHA adopted; opens R07. **Failure:** costs eat it → **document retail L/S infeasibility as a permanent finding**, route relative signals into R04’s long-only tilts, close R07. **Next:** R07 if pass; R22 either way.

R06 — Fundamental Factor Library [UPGRADE of EQUITY_CORE signals] · difficulty 5 · ~\$50/mo · 1–2 months

Backstory. Atlas has never had a fundamental datapoint in it. Value, quality/profitability, investment, issuance, accruals are among the factors that survive the big replication audits, and they diversify momentum (value-momentum negative correlation is one of the most reliable relationships in equities). Sharadar's as-reported (PIT) fundamentals make this honest at retail cost. **Mission.** Which fundamental factors replicate on retail PIT data, and do they improve R04's ranks? **Builds on:** R03 panel. **Improves:** EQUITY_CORE composite; feeds LS_ALPHA and R22's feature library. **Do.** Value composite (E/P, B/P, FCF/EV, shareholder yield), gross profitability, accruals, net issuance, asset growth, betting-against-beta. FDR control across the family — this is a factor zoo by construction, so the trial registry counts every definition tested. Sanity-check signs against published Ken-French-style series. **Success:** ≥ 3 factors with correct-sign post-2010 net spreads → blend into R04 composite (momentum+quality+value ranks). **Failure:** retail data too noisy → single-source risk documented; escalate to better data only if R22 later shows fundamentals matter. **Next:** R11, R12, R22.

R07 — Statistical Arbitrage v1 [ADD: extends LS_ALPHA] · difficulty 7 · in Norgate · 2–3 months

Backstory. Classic distance-pairs trading publicly decayed after 2010 — but the surviving literature points at residual/cluster-based variants and bear-market conditional alpha. Internally, the reversal composite's 0.407 holdout IC (measured on ETFs, never deployable there) is the repo's loudest untested lead. The ETF pair test (HYG/LQD) was two correlated ETFs, not stat arb. **Mission.** Does daily-frequency residual mean reversion pay after real costs on hundreds of stock pairs? **Builds on:** R05 (proves shorting economics first — hard prerequisite). **Improves:** LS_ALPHA with a faster, lower-beta return stream. **Do.** Cluster-based pair formation (graph clustering on correlation/sector), PCA-residual reversal portfolios, Kalman-filter hedge ratios, 1–20d holding, regime-break stops; borrow + cost accounting from R05's measured data; test the bear-market-conditional claim explicitly. **Success:** net positive expectancy at scale on 2015+ sample → adopt as LS_ALPHA fast sleeve. **Failure:** post-2010 decay confirmed for all variants → permanent close with documentation (valuable negative). **Next:** R37 (transfer-entropy pair discovery) only if this passes.

STAGE 3 — NEW MARKETS AND NEW EFFECTS

R08 — Micro-Futures Trend Platform [ADD: TREND_MACRO] · difficulty 6 · ~\$100–200/mo · 2–3 months

Backstory. Trend following is the premium with a century of evidence, positive 2020–25 out-of-sample results, and crisis convexity (2022 was a banner year) — and Atlas’s version is a long-only ETF caricature that discards the short half and the leverage. Micro futures (\$5/point on MES) made real multi-market trend replication retail-feasible; commission friction is the binding constraint below ~\$10–25k allocated. This is the Book’s first genuinely non-equity engine. **Mission.** Positive standalone expectancy, near-zero equity correlation, at solo scale. **Builds on:** R00 + owner’s capital declaration (blocking prerequisite: the futures capital band must be stated). **Adds:** TREND_MACRO slot. **Replaces:** `cta_trend_long_only` sleeve conceptually. **Do.** 10–15 micro/mini markets (MES MNQ M2K, ZN ZF, micro gold/silver/crude, micro FX majors); multi-speed TSMOM (2w–12m) + breakout ensemble; inverse-vol position scaling; 2× notional cap; roll-cost modeling; benchmark vs SG Trend index and DBMF/KMLM ETFs (if we can’t beat the ETF net of everything, hold the ETF — that is a legitimate adoption). **Success:** expectancy > 0 with corr < 0.3 to equity Book → TREND_MACRO adopted (system or ETF form). **Failure:** micro frictions eat it → hold trend exposure via ETF (DBMF/KMLM) as TREND_MACRO v0. **Next:** R09, R10, R21.

R09 — Macro Conditioning of Capital [UPGRADE of ALLOCATOR] · difficulty 5 · \$0 · 3–6 weeks

Backstory. Same evidence base as R2C, different target: instead of sub-dividing a regime state, use the growth/inflation quadrants to tilt capital *across sleeves* monthly (equity vs trend vs defensive). The literature (2024–25 TAA-with-macro-regimes work) supports exactly this monthly-cadence use. **Mission.** Do macro quadrants improve the Book’s sleeve mix over static weights? **Builds on:** R2C classifier (vintage, lagged); R08 (needs ≥2 sleeves to allocate between). **Improves:** ALLOCATOR (pre-R44 interim version). **Do.** Quadrant-conditional sleeve budgets, monthly, walk-forward; compare vs fixed mix; robustness: quadrant-boundary perturbations. **Success:** Book CAGR improves in walk-forward → adopt as interim allocator until R44. **Failure:** no allocation value → quadrants remain a REGIME_BRAIN input only (if R2C passed) or close macro entirely (if R2C also failed). **Next:** R44 inherits whatever wins.

R10 — Cross-Asset Carry [ADD: extends TREND_MACRO] · difficulty 6 · in futures data · 2–3 months

Backstory. Carry — being paid for holding the unpopular side of a curve — is the classic diversifier to trend (negatively correlated in crises, positively additive in calm). Atlas’s only carry attempt was an ETF proxy, long-only, which failed exactly as Lesson 2 predicts. Real carry lives on futures curves: backwardation/contango cross-sections in commodities, rate differentials in FX, term premium in bonds, roll-down in VIX futures. **Mission.** Positive-expectancy carry sleeve uncorrelated to the trend sleeve. **Builds on:** R08 infrastructure. **Improves:** TREND_MACRO into a trend+carry engine. **Do.** Commodity curve carry (long backwardated / short contangoed micros), FX carry+momentum+value combined (never carry alone — the crash literature is clear), bond carry, small VIX-futures roll-down sleeve with hard sizing caps. Crash-profile reporting mandatory (carry’s tails are its price). **Success:** carry adds return at low correlation to trend → merge into TREND_MACRO. **Failure:** carry crashes dominate the sample → trend-only TREND_MACRO stands. **Next:** R44.

R11 — Event-Driven Probe [PROBE→ADD: EVENT] · difficulty 6 · low cost · 2 months

Backstory. Merger arbitrage is having its best stretch since 2021 (~500bps median spread over cash; faster regulatory clocks), and it is one of the few hedge-fund staples executable manually at retail in cash deals.

Spinoffs, buybacks/issuance, and lockups have decent literatures. The disappearing index effect is the known dud — included as a negative control. **Mission.** Which event families produce per-event expectancy worth a Book slot? **Builds on:** R06 (event data via Sharadar + EDGAR parsing). **Adds:** EVENT slot if any family passes. **Do.** Backtest-then-paper: cash-deal merger arb (spread capture vs deal-break sizing), spinoff parents/children, buyback announcers, index-effect (expected negative). Position caps per deal; break-risk accounting. **Success:** merger sleeve > cash + 3pp with controlled breaks → EVENT adopted (semi-systematic is acceptable). **Failure:** spreads < retail execution → close; revisit only if deal-flow regime changes. **Next:** R44.

R12 — Earnings Alpha: PEAD + Revisions [UPGRADE of EQUITY_CORE] · difficulty 6 · low · 2 months

Backstory. Post-earnings-announcement drift is contested territory: gone from large caps since ~2006 by some accounts, alive in small/mid caps per 2025 papers. Revision momentum (analysts chasing their own errors) is steadier. Atlas never had earnings dates in its data. Under return-first this is an EQUITY_CORE feature upgrade: earnings-aware entries/exits and a small dedicated small-cap drift sleeve. **Mission.** Is there net return in post-announcement drift and revision ranks on honest PIT timing? **Builds on:** R03 + R06 (Sharadar events/estimates). **Improves:** EQUITY_CORE features; possible small EVENT sub-sleeve. **Do.** SUE ranks, announcement-window returns, drift capture 1–13 weeks, revision momentum ranks; small/mid-cap emphasis with honest liquidity-cost modeling; the large-cap null is the control. **Success:** small-cap drift net positive post-2015 → adopt as feature + sub-sleeve. **Failure:** dead everywhere → permanent close, feature set still keeps earnings dates (risk management around announcements is free).

R17 — Overnight & Daily Effects [ADD: SHORT_HORIZON] · difficulty 6 · ~\$130–200/mo · 2 months

Backstory. The overnight anomaly — close-to-open returns carrying most of equity drift, and essentially all of momentum's alpha (overnight 3-factor alpha 0.95%/mo vs 0.11% intraday) — is persistent since the 1990s, unexplained, and *structurally invisible* to a weekly-rebalance system. This family was never rejected by Atlas; it was never observable. **Mission.** Can close-to-open exposure patterns add net return after honest overnight spread/slippage? **Builds on:** R00 cost library; needs intraday/open-close data (first Polygon/Databento purchase) and an event-driven backtest engine (build here — LEAN local or custom; this engine is also R13's prerequisite quality bar). **Adds:** SHORT_HORIZON slot. **Do.** Overnight-hold sleeves on liquid ETFs/large caps; momentum-conditioned overnight exposure (hold winners overnight, flat intraday); turn-of-month and seasonality overlays as free riders; execution modeled at MOC/MOO with measured spreads; random-timing nulls. **Success:** net overnight capture > 2pp/yr on deployed capital → SHORT_HORIZON adopted. **Failure:** costs consume it (the honest risk) → close family, keep the event-driven engine as infrastructure. **Next:** R18 only if pass.

R18 — Swing Systems (2–10 day) [UPGRADE of SHORT_HORIZON] · difficulty 7 · same data · 2 months

Backstory. Between overnight and monthly lives the swing band: gap reversals, post-shock reversion, volume-spike continuation. Retail-sized, capacity-limited edges that institutions leave behind. Internally, short-horizon reversal families showed the repo's best ICs; externally the band is cost-sensitive but real. **Mission.** Positive per-trade expectancy after costs across three regimes. **Builds on:** R17 engine + panel. **Improves:** SHORT_HORIZON. **Do.** Gap-reversal entries, post-shock (>2σ) reversion, volume-conditioned continuation; per-trade stops; expectancy accounting per setup family; random-entry nulls. **Success:** ≥1 setup family with robust expectancy → add to SHORT_HORIZON. **Failure:** noise → close band, document.

R19 — Free Alt-Data Screens [FEEDER/UPGRADE] · difficulty 4 · \$0 · 3–6 weeks

Backstory. SEC EDGAR is a free, timestamped alt-data goldmine Atlas never touched: Form 4 insider cluster buys (decent literature), 13F crowding, congressional disclosures (45-day lag — mostly entertainment, test anyway as a control). All lag-honest by construction since filings carry timestamps. **Mission.** Do free filing-derived signals add anything to R03/R06 ranks? **Builds on:** R03 panel. **Improves:** EQUITY_CORE/LS_ALPHA features. **Do.** EDGAR ingestion pipeline; insider cluster-buy screens (multiple insiders, open-market, size thresholds); 13F holdings-concentration crowding measure (also feeds R39); event-study alphas with entries at filing-date+1. **Success:** insider cluster alpha positive at honest lags → feature adopted. **Failure:** all priced by filing date → close, keep the pipeline (13F crowding still feeds R39).

R20 — LLM Text Signals with PIT Discipline [UPGRADE of feature library] · difficulty 7 · low-med · 2–3 months

Backstory. 2024–25 literature: LLM-extracted news/filings sentiment shows 3–10 day predictive windows; the two failure modes are backfill bias (news archives rewritten after the fact) and priced-in-ness (signal real but gone before you can trade). The filings-change effect (“Lazy Prices”: firms that quietly rewrite their 10-K underperform) is slower and free to test on EDGAR. **Mission.** Do text features add IC over the price/fundamental feature set *at tradable lags*? **Builds on:** R03 panel + R19 EDGAR pipeline. **Improves:** feature library for R22. **Do.** Start with filings diffs (free, timestamped): year-over-year 10-K/Q similarity scores. Then transcript tone/uncertainty (cheap APIs, timestamp-audited). Then news sentiment only with a verified PIT news source. Mandatory control: enter at t+2 — if all signal lives at t0, it was never tradable. LLM outputs cached and versioned for reproducibility. **Success:** any text feature adds net IC at t+2 → adopt into R22’s library. **Failure:** all priced-in → close news branch; filings-diff may still survive as slow factor.

R21 — Crypto Sleeve Probe [PROBE] · difficulty 4 · \$0 · 2–4 weeks

Backstory. BTC/ETH trend-follows well historically (retail-driven, high-vol trends), funding/basis carry exists but compressed (basis $_{25\%} \rightarrow 4.5\%$ over 2024–25; funding positive ~92% of the time on majors), and US-person access to offshore perps is a hard constraint. Worth exactly one bounded probe at small risk budget — a return-first program shouldn’t ignore the most volatile liquid asset class, or over-commit to it. **Mission.** Does BTC/ETH trend + regulated-venue basis justify a 5–10% risk-budget sleeve? **Builds on:** R08 trend code (same TSMOM machinery, new asset). **Adds:** small TREND_MACRO extension if pass. **Do.** TSMOM ensemble on BTC/ETH (free exchange data); regulated basis capture feasibility note (CME micro BTC futures vs spot); venue-risk assessment. **Success:** positive expectancy + diversification at acceptable venue risk → small sleeve. **Failure:** whipsaw-dominated post-2022 → close; re-probe annually (R63).

STAGE 4 — OPTIONS: THE UNTESTED DOMAIN

R13 — Options Data Foundation + Vol Forecasting [FEEDER] · difficulty 6 · ~\$100–160/mo · 2 months

Backstory. Lesson 5: every internal options verdict priced a Black–Scholes fiction. Before any options strategy, we need real chains (ORATS/ThetaData, 2007+) and a forecasting layer: HAR-family realized-vol models (the workhorse of the vol literature — beats GARCH), implied-vs-realized gap tracking, term/skew features, and GEX-style dealer-positioning proxies (popular, mixed formal evidence — test against nulls). **Mission.** Can we forecast realized vol and identify when implied is rich/cheap, better than unconditional rules? **Builds on:** R00; the R17 event-driven engine if built (otherwise build the options simulator here). **Feeds:** R14, R15, R16, and vol-features into R22. **Do.** Ingest chains; build surface features; HAR + implied blend forecasts; conditional VRP sign prediction (when does short premium lose?); arbitrage-free surface sanity checks; dealer-positioning features with strict null controls. **Success:** conditional edge over unconditional VRP established → R14/R15 proceed with signals. **Failure:** no conditioning edge → R14 still tests *unconditional* defined-risk premium (evidence says it pays on average), R15 still tests recovery convexity.

R14 — Defined-Risk Volatility Premium [ADD: OPTIONS income leg] · difficulty 7 · same · 2–3 months

Backstory. Selling insurance pays: the variance risk premium is one of the most persistent premia known — and it blows up (2018 –90%+ for naked products, 2020, the 2025 tariff spike: SVOL –33%). Crowding halved after 2018 (VIX futures OI at half pre-COVID), arguably improving the premium. The retail-survivable expression is *defined-risk only*: put credit spreads and iron condors where max loss is structural, plus small VIX term-structure carry. Atlas's regime engine — which correctly identifies stress — is a genuine comparative advantage for *when not to sell*. **Mission.** Net CAGR contribution from premium selling that survives its own tail episodes. **Builds on:** R13 data/signals + REGIME_BRAIN gating (sell premium only outside stress/pre-stress states — pre-register the gate as the *existing* regime states, unchanged, to avoid hindsight gate design). **Adds:** OPTIONS income leg. **Do.** SPY/SPX put credit spreads and condors, 2–8 week tenors; sizing so a max-loss month costs ≤2% of Book; VIX carry sleeve capped small; stress replay of 2018/2020/2022/2025 analogs including gap-through-strike scenarios. **Success:** +2pp CAGR contribution surviving all replayed episodes → adopt. **Failure:** tails erase multi-year premium even defined-risk → close income leg; R15's long-convexity leg is unaffected.

R15 — Recovery Convexity on Real Chains [ADD: OPTIONS convexity leg] · difficulty 6 · same · 2 months

Backstory. The one internal options survivor: long calls bought only at defensive→risk-on recovery inflections beat baseline even on pessimistic proxy pricing (0.958 vs 0.948) — but only barely beat an equivalent delta-one tilt (+0.008), on a premium budget so tiny (0.5% NAV) it couldn't matter. R02's PBI state and this run are the same thesis at different convexity: when improving-panic confirms, upside is violent (+73% annualized episodes) and implied vol is still elevated-but-falling — the textbook setup for call spreads. **Mission.** Does real-chain convexity at PBI/recovery triggers beat the equivalent ETF tilt by enough to earn the slot? **Builds on:** R02 (triggers) + R13 (chains). **Adds:** OPTIONS convexity leg. **Replaces-if-better:** pure delta-one re-risking from R02 at episode starts. **Do.** Call spreads/outright calls on SPY/QQQ at PBI fire dates; DTE 60–120; premium budget grid 1/2/4% NAV per episode; the *tilt control* (same signal, same dollars, delta-one) is mandatory — options must beat it, not just baseline; real IV entry levels from chains decide spread-vs-outright per episode. **Success:** options > tilt meaningfully on real data → convexity leg adopted for recovery episodes. **Failure:** tilt equivalence → drop options leg, R02 re-risks with shares, savings on theta documented.

R16 — Dispersion & Exotic Structures: Feasibility Note [PROBE] · difficulty 9 · same · 2–3 weeks

Backstory. Dispersion (short index vol vs long single-name vol) is institutionally lucrative and retail-hostile: the edge dies in execution across hundreds of single-name spreads. One honest feasibility note prevents this idea from haunting future sessions. **Mission.** Is any simplified expression (5–10 name “dirty dispersion”) viable after full spread costs? **Builds on:** R13. **Do:** paper-model a 5-name dirty dispersion basket at real spreads; document the execution arithmetic. **Success:** (unlikely) viable → small experiment. **Failure:** documented close — Tier-6 status confirmed, never revisit without institutional execution.

STAGE 5 — THE ML LADDER (SIGNAL FACTORY ON THE PANEL)

R22 — Gradient-Boosted Ranking Engine [UPGRADE of EQUITY_CORE/LS_ALPHA signals] · difficulty 6 · GPU optional · 2–3 months

Backstory. Lesson 4 said models can't beat features on 1,000 samples. On a panel of 1,000 stocks $\times$ 3,000 days, the evidence flips: gradient-boosted trees and shallow nets roughly *double* linear strategies' performance in the canonical study and its replications. This is the single most evidence-backed ML bet available, and everything Atlas learned about leakage control now gets applied at proper scale. **Mission.** Beat the linear composite rank (R03/R06 blend) by $\geq 50\%$ in IC and in net top-k return. **Builds on:** R03 panel, R06 factors, R19/R20 features — the 150+ feature library. **Improves:** the ranking signal feeding R04/R05. The linear composite stays as the permanent baseline and fallback. **Do.** LightGBM ranker; purged K-fold with embargo; expanding-window walk-forward; DSR/PBO per experiment family; SHAP audits (a model whose top features make no economic sense is presumed leaky until proven otherwise); seed-stability (5 seeds, report dispersion). **Success:** GBM > linear by pre-registered margin OOS $\rightarrow$ EQUITY_CORE/LS_ALPHA signals upgraded. **Failure:** parity $\rightarrow$ linear wins (cheaper, robust); ladder above shrinks to curiosity budget. **Next:** R23 always; R24+ only on success.

R23 — Targets & Labels Laboratory [FEEDER] · difficulty 5 · existing · 3–6 weeks

Backstory. Internal work (triple-barrier, decision labels) already hinted that *what you predict* matters more than *what you predict with*. The literature agrees: rank targets, threshold targets ("will it beat costs?"), and barrier labels each beat naive point-return regression in different settings; meta-labeling (predicting when your own signal works) adds a cheap layer. **Mission.** Which target maximizes realized top-k return per unit of overfit risk, on identical features/models? **Builds on:** R22 setup. **Improves:** every learning run after it. **Do.** Fixed GBM + features; swap targets: forward return, rank, top-quintile membership, P(beat costs), triple-barrier, MFE/MAE, log-growth contribution; meta-label layer on the best; identical validation everywhere. **Success:** a target dominates $\rightarrow$ becomes house standard. **Failure:** indifference $\rightarrow$ simplest target (rank) wins by default.

R24–R29 — The Challenger Ladder [REPLACE-IF-BETTER vs R22] · difficulty 5–8 · GPU rental · interleaved

Backstory. Five model families with loud literatures and thin honest finance evidence. Each gets one controlled shot at the R22 champion — same features, same validation, same trial accounting. The published pattern to expect: temporal CNNs (R24) and LSTM/S4/Mamba (R25) rarely beat tuned GBMs on tabular financial panels; transformers/cross-asset attention (R26) occasionally add value on relational structure; GNNs (R27) mostly re-discover what graph-derived *features* already give a GBM — so the baseline is GBM+graph-features, not vanilla GBM; time-series foundation models (R29 — TimesFM/Chronos zero-shot and fine-tuned) showed near-parity with scratch baselines on equities in 2025 benchmarking. R28 (self-supervised embeddings, conditional autoencoders) doubles as the feature-refresh for R30. **Mission (each):** beat the reigning champion OOS with seed stability, or close the branch cheaply. **Builds on:** R22/R23. **Rule:** losers are documented and closed — a confirmed negative here saves months later. Only a winner replaces the champion signal. **Success/Failure:** per-rung pre-registered margins; any rung that wins gets a locked confirmation before adoption (the α /PBI pattern).

R30 — Analog Retrieval v2 [UPGRADE of REGIME_BRAIN] · difficulty 5 · existing · 3–6 weeks

Backstory. The internal kNN analog engine was Atlas's most honest ML result: real signal, matched the hand rule, features exhausted. The features are no longer exhausted — R01 breadth, R2C macro factors, R13 vol-

surface features triple the state space. Retrieval (“what happened after the 25 most similar market states?”) is interpretable and cheap. **Mission.** Does enriched-feature retrieval finally beat the hand-built state-quality composite? **Builds on:** R01+R2C+R13 features; internal M2 code. **Improves-Replaces:** R2A composite inside REGIME_BRAIN if it wins. **Do.** Re-run the M2 design (k=25, embargo, expanding standardization) on the enriched state vector; same nulls as Moonshot; portfolio-value comparison, not just IC. **Success:** beats hand composite in decision value → REGIME_BRAIN input upgraded. **Failure:** ceiling persists → strong evidence the regime problem is feature-complete; stop spending on it.

R31 — Decision-Focused Learning [REPLACE-IF-BETTER vs predict-then-rank] · difficulty 8 · GPU · 2–3 months

Backstory. 2024–25 literature: training the forecaster through the portfolio objective (differentiable optimization layers, Sharpe/growth losses) beats two-stage predict-then-optimize on decision metrics — with a documented pathology of turnover inflation. Atlas’s Frontier P6 built decision labels but never closed the loop. **Mission.** Does end-to-end training of the top-k allocation beat R22’s two-stage pipeline *net of the extra turnover*? **Builds on:** R22 champion as baseline. **Do:** cvxpylayers/SPO+ top-k layer; log-growth and cost-aware losses; turnover-inflation check pre-registered as the primary failure mode. **Success:** net improvement → allocation head adopted. **Failure:** turnover eats it → close, keep two-stage.

R32–R34 — Execution RL, Generative Stress, Automated Discovery [bounded frontier]

R32 Execution RL (after live paper fills exist, R46): offline RL on our own order data vs TWAP/limit-ladder rules; scope strictly execution — portfolio RL stays banned (OOS failure record + sample arithmetic). Success = implementation-shortfall reduction $\geq 20\%$. **R33 Generative stress engine** (after R44): TimeGAN/diffusion scenario generation for stress-testing the Book — never for alpha training (that use is the negative control). Success = coverage of 2008/2020-type paths the historical record undersamples. **R34 Automated discovery with guardrails:** LLM hypothesis agents + genetic feature synthesis, allowed only with: every generated candidate logged, family-wise DSR, MDL complexity penalties, and a standing rule that nothing agent-found is believed without a locked confirmation run. Mission: one novel validated signal per quarter without becoming an overfit factory.

STAGE 6 — MATHEMATICS TRANSFERS (BOUNDED EXPERIMENTS)

R35 — MPC Rebalancing [UPGRADE of Book plumbing] · difficulty 6

Backstory. Model-predictive control (receding-horizon optimization with costs) is the established math for multi-period rebalancing (Boyd's *cvxportfolio* lineage). Atlas rebalances myopically. Costs are the tax every sleeve pays; cutting them is certain return. **Mission/Success:** turnover down $\geq 20\%$ at flat-or-better return across the Book → adopt as standard rebalancer.

R36 — Spectral & Signature Features [FEEDER probe] · difficulty 6

Backstory. Wavelet multi-scale decomposition and path signatures (active 2024–25 finance literature) are principled ways to encode path shape that hand features may miss. Boundary leakage is the classic implementation bug — audit for it explicitly. **Mission/Success:** incremental IC in R22 ablations → features adopted; else closed.

R37 — Transfer-Entropy Lead-Lag [PROBE] · difficulty 7

Backstory. Atlas's weekly ETF lead-lag failed (wrong horizon, wrong universe). Information-theoretic directed-flow measures on *daily stock/sector* data are the honest retest, with brutal multiple-testing control. **Mission/Success:** TE-discovered pairs beat correlation-discovered pairs in R07's machinery → feeds LS_ALPHA; else closed. Requires R07 alive.

R38 — Geometry Suite: OT, DRO, TDA [PROBE, negative-control framing] · difficulty 7

Backstory. Optimal-transport regime distance and distributionally-robust allocation have credible literatures; persistent-homology crash prediction mostly proxies correlation/vol (expected negative — include as the control that keeps us honest). **Mission/Success:** any component beats its simple baseline (z-score regimes; mean-variance; vol-based warning) → adopt that component; publish the negatives.

R39 — Network Crowding Gauge [REPLACE-IF-BETTER vs leadership guard] · difficulty 6

Backstory. The production fragility guard uses ETF leadership as a crowding proxy — it works but costs SPY exposure. 13F co-ownership centrality (free data via R19) is a richer crowding measure. Internal warning encoded: the absorption-ratio throttle failed because fragility de-risking double-counts stress on a *defensive* base — test only on the offensive Book. **Mission/Success:** crowding-conditional performance beats the leadership guard → guard replaced; else keep.

R40 — Causal Robustness Screens [FEEDER] · difficulty 7

Backstory. The decay problem (factors fade post-publication) is a causality problem: signals that survive orthogonalization (double-ML) and stay stable across regime environments (invariant prediction) should decay slower. If the screen predicts OOS survival, it becomes a standing filter on every signal the factory produces. **Mission/Success:** invariant signals demonstrably decay less in our own registry data → screen institutionalized.

R41 — Complex-Systems Early Warning [PROBE, expected modest] · difficulty 7

Backstory. Critical-slowing-down/contagion indicators claim crash early-warning. High false-positive rates documented. One bounded test against VIX/breadth baselines; feeds REGIME_BRAIN only if it genuinely adds warning time.

STAGE 7 — INTEGRATION: TURN ENGINES INTO A MACHINE

R42 — Sizing Laboratory [ADD: SIZING] · difficulty 5 · prerequisite: ≥3 adopted engines

Backstory. Atlas sized by inverse-vol and Sharpe-first instinct. Return-first sizing is Kelly mathematics: log-growth-optimal fractions, shrunk for estimation error (half-Kelly as the sane default), conviction-scaled by signal strength. Sizing is the highest-leverage free upgrade in the program — the same signals at better sizes compound visibly faster. **Mission.** The sizing rule that maximizes Book log growth at survivable (recorded, not gated) drawdown. **Builds on:** every adopted engine's return series. **Adds:** SIZING slot. **Do.** Compare equal-risk, inverse-vol, fractional Kelly ($\frac{1}{4}/\frac{1}{2}/\text{full}$), conviction-scaled, conformal-interval-scaled; estimation-error stress (bootstrap the inputs); report growth-vs-drawdown frontier. **Success:** growth-optimal sizing beats naive at same realized drawdown → adopt. **Next:** R43.

R43 — Dynamic Leverage $\leq 1.5\times$ [UPGRADE of SIZING] · difficulty 6 · prerequisite: owner capital declaration

Backstory. The Book runs unlevered until here by design. Modest leverage on a diversified multi-engine book is the classic final return lever: financing at IBKR tiered rates (or embedded via futures), never daily-reset levered ETFs (decay documented). Leverage amplifies model error exactly when correlations converge — hence last, capped, and state-aware. **Mission.** Does $1.0\text{--}1.5\times$ state-conditional leverage raise compound growth net of financing? **Builds on:** R42 sizing + REGIME_BRAIN (de-lever in stress states — the one defensive reflex allowed early because it's return-motivated: avoiding ruin is a growth strategy). **Success:** levered log growth > unlevered with survivable stress replay → adopt with hard $1.5\times$ cap. **Failure:** financing + error erase it → Book stays unlevered; futures notional remains the only leverage.

R44 — Multi-Strategy Capital Allocation [ADD: ALLOCATOR] · difficulty 6 · prerequisite: ≥3 engines in paper

Backstory. The end-state: capital flowing across genuinely different engines (equity, trend, options, events, short-horizon) — the diversification Atlas faked with eight flavors of long-only equity beta. Combining modestly-good uncorrelated engines beats perfecting any single one; that arithmetic is the whole reason multi-strategy funds exist. **Mission.** Book-level net CAGR above the best single engine. **Builds on:** all adopted engines; R09's interim allocator; RMT-cleaned covariance; strategy-momentum tilts (allocate more to what's working, slowly). **Do.** Risk-budgeted baseline; monthly reallocation; correlation-regime awareness (crisis convergence modeled); drawdown-aware de-allocation *rules recorded but not yet binding* (they bind in R45). **Success:** Book > best engine → ALLOCATOR adopted; this Book is the machine. **Failure:** engines too correlated → the honest finding that we built one engine several ways; concentrate capital in the best and keep hunting (Stage 8).

R45 — Drawdown Engineering, Priced [UPGRADE of Book — the deferred metrics arrive] · difficulty 5

Backstory. Now — with real alpha assembled — the other metrics get their turn, exactly as the mandate ordered: last. Every protection (per-position stops, regime de-risking, vol caps, R15 tail hedges, CVaR budgets) is added only with its price tag measured: CAGR given up per point of drawdown saved. Budget: max 20% of Book CAGR may be spent on protection. The old Atlas (DEFENSE slot) competes here as one protection option among several — its true cost is now measurable. **Mission.** Cut worst-case damage meaningfully for $\leq 20\%$ CAGR give-up. **Success:** MaxDD down $\geq 30\%$ within budget → protections adopted. **Failure:** protection costs more than it saves → run the Book rawer, size DEFENSE slot accordingly, document.

R46 — Execution & Cost Engine [FEEDER, continuous from first paper trade]

Backstory. The 10bps flat assumption once gated real signal amplitude (the α story). From the first paper order onward, measure actual spreads/fills, build limit-order policies, feed measured costs back into every backtest. Cost cutting is riskless return. **Mission/Success:** measured costs $\leq$ modeled; models continuously updated; opens R32 if rule-based execution leaves shortfall on the table.

R47 — Paper-Trading Gate [the only gate before real money] · difficulty 4 · 13+ weeks minimum

Backstory. Nothing in this project has ever traded live; there is no internal prior on backtest-to-live degradation. The paper period builds it. Human authorization required for any real capital — that rule survives from the old governance untouched. **Mission.** Book tracks its backtest expectation within declared confidence bands for 13+ consecutive weeks. **Success:** owner decision on live capital with degradation data in hand. **Failure:** degradation beyond bands → back to the responsible run with live-vs-backtest diagnostics (usually costs → R46, or regime → R02-family).

STAGE 8 — EVERGREEN RUNS: THE PROGRAM THAT NEVER RUNS OUT

These recur or unlock conditionally — the standing supply of next runs.

R48 — Quarterly Signal Refresh [recurring]

Re-validate every adopted signal on the newest quarter: IC drift, decay curves vs R40's predictions, regime-conditional health. Any signal two consecutive quarters below its adoption threshold goes to the bench; benched signals showing revival get one locked re-confirmation. *This run is the Book's immune system.*

R49 — Universe Expansion: Mid/Small Caps [conditional on R03]

Extend the panel to Russell-2000 names with honest liquidity costs. Momentum/reversal/PEAD effects are historically stronger down-cap; costs are too. Return-first question: does the extra spread survive the extra friction?

R50 — International via ADRs [conditional on R03]

Rerun the R03/R06 factory on liquid ADRs: fresh cross-section, different macro exposures, same infrastructure. Also the honest out-of-sample test for every US-discovered signal.

R51 — Sector & Industry Rotation Revisited [conditional on R01]

Old sector rotation failed on ETF breadth; with stock-level breadth per sector (R01 data), rebuild sector scores bottom-up. Replaces the failed Phase 4B approach if it wins.

R52 — Ensemble Regeneration [recurring, semi-annual]

Re-blend all live signals with fresh IC-weighted / regime-conditional weights (the R22 champion re-trained, the linear fallback re-fit). Guards against silent staleness of the combination layer even when components stay healthy.

R53 — Bench Resurrection Sprint [recurring, annual]

Everything ever benched (failed runs, retired signals, losing challengers) gets one annual look: has new data/infrastructure changed any verdict? The registry's `no_retest_justified` rows stay dead; everything else is eligible. This is where “completely replace a part” happens on schedule.

R54 — Capacity & Slippage Audit [recurring, tied to capital growth]

As capital grows, re-test every sleeve's edge at the new size; retire capacity-constrained edges gracefully (swing/event sleeves first).

R55 — New-Data Probe Slots [standing]

A permanent budget for one new dataset probe per quarter (next candidates: cheap estimate feeds for R12 upgrade, borrow-free history vendors, transcript embeddings archive, futures tick data for R08 speed-up). Each probe: 2 weeks, pre-registered, adopt-or-drop.

R56 — Volatility Surface Alpha [conditional on R13/R14 adoption]

Second-generation options: cross-sectional single-name vol screens (rich/cheap IV vs forecast), earnings-vol calendars, skew trades. Only if the index-level program earned its slot.

R57 — Treasury & Rates Macro Sleeve [conditional on R08]

Duration timing and curve trades (2s10s steepeners/flatteners via micro yield futures) using R2C's macro factors. The natural second use of the futures stack.

R58 — Leveraged Recovery Protocol [conditional on R02+R43]

The maximum-aggression composite: PBI-confirmed early recovery + R43 leverage + R15 convexity simultaneously — the full return-first expression of the single best episode type in the dataset. Pre-registered episode caps; this is the Book's designated "swing big when it's time" module.

R59 — Short-Side Alpha Specialization [conditional on R05]

If shorting economics work, build the dedicated short book: issuance/dilution screens, accrual deterioration, failed-momentum, lockup expiries. Short alpha decays slower (fewer players can hold it).

R60 — Cross-Engine Signal Sharing [conditional on ≥ 3 engines]

Systematically test every engine's internal state as a feature for every other engine (trend-sleeve drawdown as equity de-risk signal; options skew as reversal timing; breadth as futures filter). The correlation structure of *signals*, not just returns.

R61 — Second ML Generation [conditional on R22 + 2 years of data growth]

Re-run the challenger ladder (R24–R29 families, updated architectures) once the panel has grown and the first generation's verdicts are two years stale. Model verdicts expire; this run enforces that.

R62 — Alternative Return Streams Probe [standing, annual]

One structured look per year at anything newly feasible: prediction markets at scale, new listed products, tokenized assets with real liquidity, new micro futures launches. Two-week probes, registry-logged.

R63 — Annual Program Audit [recurring]

The meta-run: re-read this run book against the registry; retire finished branches; write the next year's run cards; re-rank everything by realized (not projected) return contribution. Produces version N+1 of this document.

PART D — QUICK REFERENCE

D.1 Run index by Book component

Component	Creating run	Upgrading runs	Potential replacers
EQUITY_CORE	R03/R04	R06, R12, R19, R20, R22, R23, R49–R52	R31 (allocation head)
REGIME_BRAIN	R02	R01, R2B, R2C, R30, R41	R30 (if retrieval wins)
LS_ALPHA	R05	R07, R22, R37, R59	—
TREND_MACRO	R08	R10, R21, R57	ETF fallback (DBMF/KMLM)
OPTIONS	R14/R15	R13, R56	R15 tilt-control may delete convexity leg
SHORT_HORIZON	R17	R18	—
EVENT	R11	R12	—
SIZING	R42	R43, R58	—
ALLOCATOR	R09 (interim)	R44, R35, R60	R44 replaces R09
DEFENSE	legacy Atlas	R45 prices it	R45 may shrink it to zero

D.2 If you only have time for five runs

R01 (breadth) → R02 (PBI native) → R03 (stock universe) → R04 (Book v1) → R08 (futures trend). That sequence alone plausibly moves the Book from a 7% defensive machine to a diversified return-seeking one, for ~\$250/month of data.

D.3 Failure is output

Every closed branch above is listed with its closure condition. A run that ends in a documented, null-controlled negative has done its job — it prevents the same idea from consuming a future session. The registry (`atlas_offensive_future_run_registry.csv` + trial registry from R00) is the memory; this book is the map. Update both, always.

Master Run Book v1.0 · 2026-07-21 · Companion to the Atlas Offensive Exhaustive Discovery Blueprint. The blueprint holds the full evidence base (catalogs, ledgers, coverage disclosure); this book holds the executable program. Production pin untouched: `improved_frontier_phase5_fragility_guard`.